

Sri Lanka Institute of Information Technology



STOCK MARKET DIRECTION PREDICTION

Artificial Intelligence and Machine Learning Project Documentation
2026-Y2-S1-MLB-B1G2-06

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Declaration and AI Tool Usage

We confirm that this report documents the work performed for the assigned stock-market dataset. The preprocessing, model training, validation, testing and comparison outputs were executed and checked by the project group. Results were retained even when performance was weak, and no metric was altered or fabricated.

ChatGPT was used to support explanations, code development, debugging and documentation. Google Colab and Python libraries were used to execute the workflow, create visualizations, train models and calculate evaluation metrics. The group remains responsible for understanding, validating and presenting the submitted work.

Group Members and Contributions

Student ID	Student Name	Main Contribution
IT25102978	Jayasundara J.M.Y.V.	Missing data handling and Logistic Regression
IT25102165	Halangoda R.W.W.M.M.C.	Categorical encoding and Support Vector Machine
IT25200808	Shavindi T.D.P.	Outlier handling and Decision Tree
IT25101173	Randil B.R.K.	Feature scaling and K Nearest Neighbors
IT25200151	Wanniarchchi W.K.A.N.	Feature engineering and Random Forest
IT25102034	Heshan I.A.M.	Feature selection and Gradient Boosting

Executive Summary

This project investigates whether classical machine-learning models can predict the direction of the next closing price for international stock-market indices. The task was formulated as binary classification: Class 1 represents an increase in the next closing price, while Class 0 represents a decrease or no increase.

The original dataset contained 112,457 daily records across 14 market indices. After validation, cleaning and target construction, 110,223 records remained. A chronological split produced 77,150 training records, 16,534 validation records and 16,539 test records. The integrated pipeline handled missing data, invalid records, outliers, categorical encoding, scaling, feature engineering and feature selection without fitting preprocessing steps on future test data.

Six supervised classifiers were implemented: Logistic Regression, Support Vector Machine, Decision Tree, K Nearest Neighbors, Random Forest and Gradient Boosting. Each experiment included baseline training, five-fold cross-validation, hyperparameter tuning, validation comparison and final testing.

K Nearest Neighbors was selected as the most meaningful observed model because it predicted both target classes in substantial quantities and achieved the highest test ROC AUC and balanced accuracy. However, its ROC AUC remained close to 0.50, indicating weak discrimination. The main conclusion is therefore not that stock direction was predicted reliably, but that the selected five-feature representation contained limited information for next-day direction prediction.

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1 Dataset Understanding and Problem Framing

1.1 Real World Problem Domain

The project belongs to the finance and stock-market analytics domain. Its objective is to predict whether the following closing price for a market index will increase. A direction forecast may support market monitoring and risk analysis, but it must not be interpreted as guaranteed investment advice because price movements depend on news, macroeconomic conditions, investor behaviour and other information not present in the dataset.

1.2 Problem Definition

For each market index, the target was constructed by comparing the following closing price with the current closing price. A value of 1 was assigned when the following close was higher; otherwise, a value of 0 was assigned. This created a supervised binary-classification task.

Target Value	Meaning
0	Next closing price decreased or remained unchanged
1	Next closing price increased

1.3 Dataset Description and Suitability

Dataset Property	Observed Value
Original size	112,457 rows and 8 source columns
Market coverage	14 international market indices
Source attributes	Date, Open, High, Low, Close, Adjusted Close, Volume and Index
Rows with missing required values	2,219 rows or approximately 1.97 percent
Exact duplicates	0
Invalid OHLC records removed	1
Final cleaned size	110,223 rows and 10 columns including engineered fields
Class 0 records	51,715
Class 1 records	58,508

The dataset is suitable because it contains a large number of chronological market observations, a clear categorical index identifier, numerical market variables and enough records to train and compare several classical machine-learning models. Its limitations include missing values, unusual volume reporting, market noise and the absence of external news or economic variables.

2 Data Preprocessing and Exploratory Analysis

2.1 Integrated Preprocessing Flow

9. Validated required columns and converted the Date field to a consistent date type.
10. Removed incomplete and invalid financial records and confirmed that no exact duplicates existed.
11. Constructed the next-day direction target separately within each market index.
12. Engineered calendar features including year, month, day and day of week.
13. Created chronological 70 percent training, 15 percent validation and 15 percent test partitions within each index.
14. Learned IQR outlier limits from training data and applied those limits consistently to all partitions.
15. One-hot encoded the market-index category and aligned the resulting columns.
16. Fitted StandardScaler only on training data and transformed validation and test data with the same scaler.
17. Fitted SelectKBest only on training data and retained the five highest-scoring predictors.

2.2 Final Selected Features

Feature	Meaning
Volume_Available	Binary indicator showing whether a positive volume value was reported
Day	Standardized calendar day of the month
Index_399001.SZ	One-hot indicator for the Shenzhen Component Index
Index_IXIC	One-hot indicator for the NASDAQ Composite Index
Index_TWII	One-hot indicator for the Taiwan Weighted Index

2.3 Data Splitting and Leakage Prevention

Partition	Records	Purpose
Training	77,150	Fit preprocessing steps and models
Validation	16,534	Compare baseline and tuned configurations
Testing	16,539	One final unbiased evaluation

A chronological split was preferred to a random split because market data are time ordered. Scalers, outlier thresholds and feature selectors were fitted using training data only. This prevents information from later observations leaking into earlier model development.

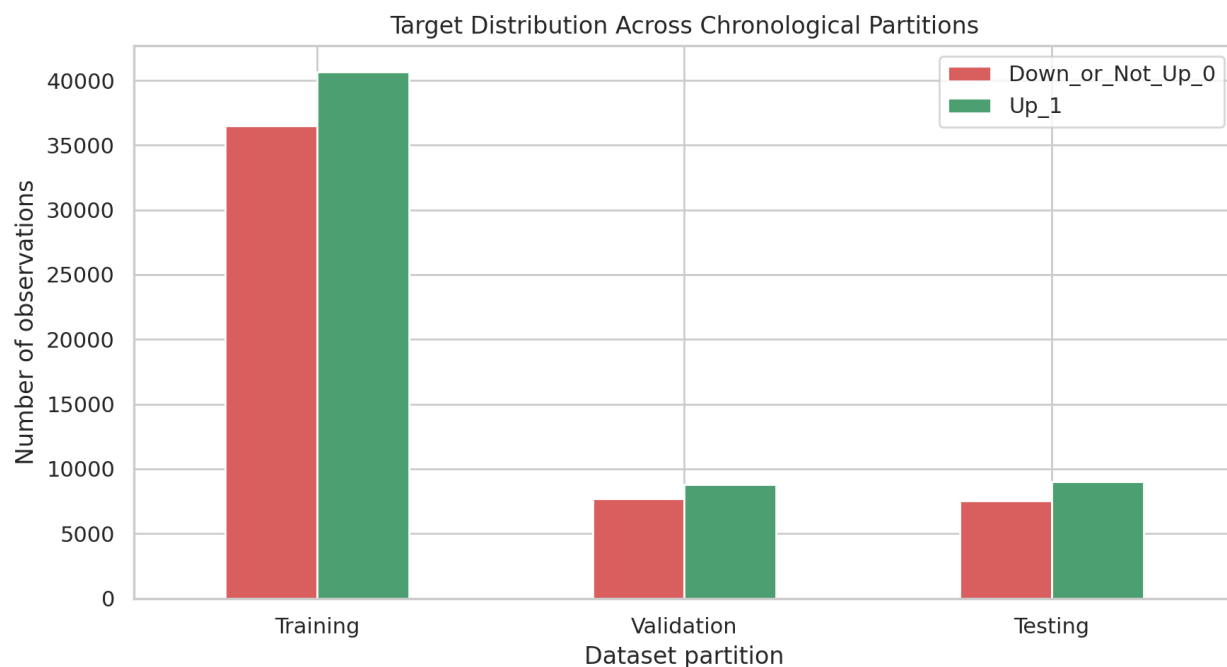


Figure 1 Target distribution across chronological dataset partitions

2.4 Exploratory Data Analysis Insights

- Class 1 was moderately more common than Class 0, and its proportion increased to 54.56 percent in the test partition.
- The majority-class test baseline was therefore 54.56 percent. Any model near this accuracy required additional examination through confusion matrices and balanced metrics.
- About 1.97 percent of source rows contained missing required values, while no exact duplicates were found.
- Zero volume occurred frequently and may indicate an index that did not report volume; it was represented through Volume_Available rather than treated automatically as an error.
- The selected features had statistically detectable but weak relationships with the target, anticipating limited model discrimination.

3 Model Design and Implementation

3.1 Selected Algorithms

Model	Justification
Logistic Regression	Interpretable probabilistic baseline for a binary target
Linear SVM	Margin-based classifier suitable for scaled numerical features
Decision Tree	Interpretable non-linear rules and feature importance
K Nearest Neighbors	Non-parametric classifier for local neighbourhood patterns
Random Forest	Bagged tree ensemble intended to reduce single-tree variance
Gradient Boosting	Sequential tree ensemble that corrects earlier errors

Using six models exceeded the requirement to implement at least two suitable AI or machine-learning algorithms. All six were evaluated on identical processed data to ensure a fair comparison.

3.2 Common Experimental Procedure

- Train a baseline model on the training partition.
- Perform five-fold stratified cross-validation.
- Tune relevant hyperparameters with GridSearchCV.
- Compare baseline and tuned variants on validation data.
- Refit the selected configuration using training and validation data.
- Evaluate once on the untouched test partition.
- Save the fitted model, predictions, metrics, reports and plots.

3.3 Best Hyperparameters

Model	Selected Parameters	Best CV F1
Logistic Regression	C: 0.01, class_weight: null, penalty: l1, solver: liblinear	0.6904
SVM	C: 0.001, class_weight: null, loss: hinge	0.6904
Decision Tree	class_weight: null, criterion: gini, max_depth: 3, min_samples_leaf: 1	0.6904
KNN	n_neighbors: 31, p: 1, weights: distance	0.5843
Random Forest	max_depth: 5, max_features: sqrt, min_samples_leaf: 1, n_estimators: 200	0.6839
Gradient Boosting	learning_rate: 0.05, max_depth: 1, n_estimators: 50	0.6904

4 Model Evaluation and Comparison

4.1 Evaluation Metrics

Metric	Interpretation
Accuracy	Proportion of all test predictions that were correct
Precision	Proportion of predicted upward movements that were actually upward
Recall	Proportion of actual upward movements detected by the model
F1 Score	Harmonic mean of precision and recall
ROC AUC	Ability to rank Class 1 above Class 0 across thresholds
Specificity	Proportion of Class 0 observations correctly detected
Balanced Accuracy	Average recall across both target classes
MCC	Correlation-style measure using all confusion-matrix outcomes

4.2 Final Test Results

Model	Accuracy	Precision	Recall	F1	ROC AUC	Both Classes
KNN	50.70%	54.78%	55.31%	55.04%	0.5086	Yes
Decision Tree	54.56%	54.56%	100.00%	70.60%	0.5069	No
Random Forest	54.54%	54.56%	99.91%	70.58%	0.5051	Yes
Gradient Boosting	54.56%	54.56%	100.00%	70.60%	0.5042	No
Logistic Regression	54.56%	54.56%	100.00%	70.60%	0.4950	No
SVM	54.56%	54.56%	100.00%	70.60%	0.4907	No

The test majority-class baseline was 54.56 percent. Logistic Regression, SVM, Decision Tree and Gradient Boosting reached approximately this accuracy by predicting every test record as Class 1. Random Forest predicted only 13 records as Class 0. These results show why accuracy and F1 score cannot be interpreted alone.

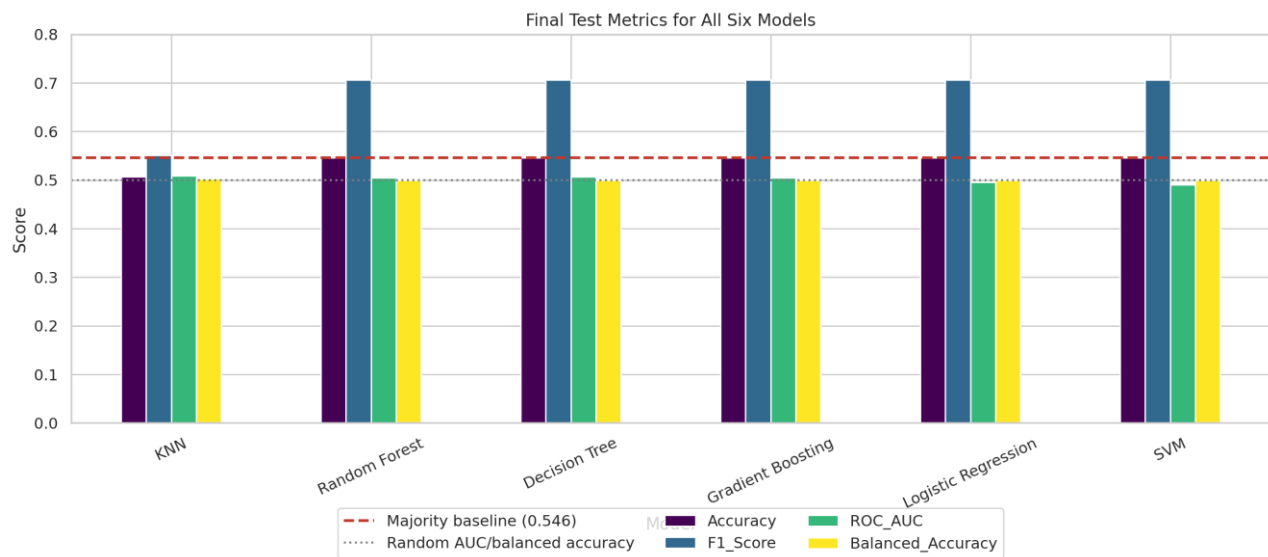


Figure 2 Final test metric comparison for all six models

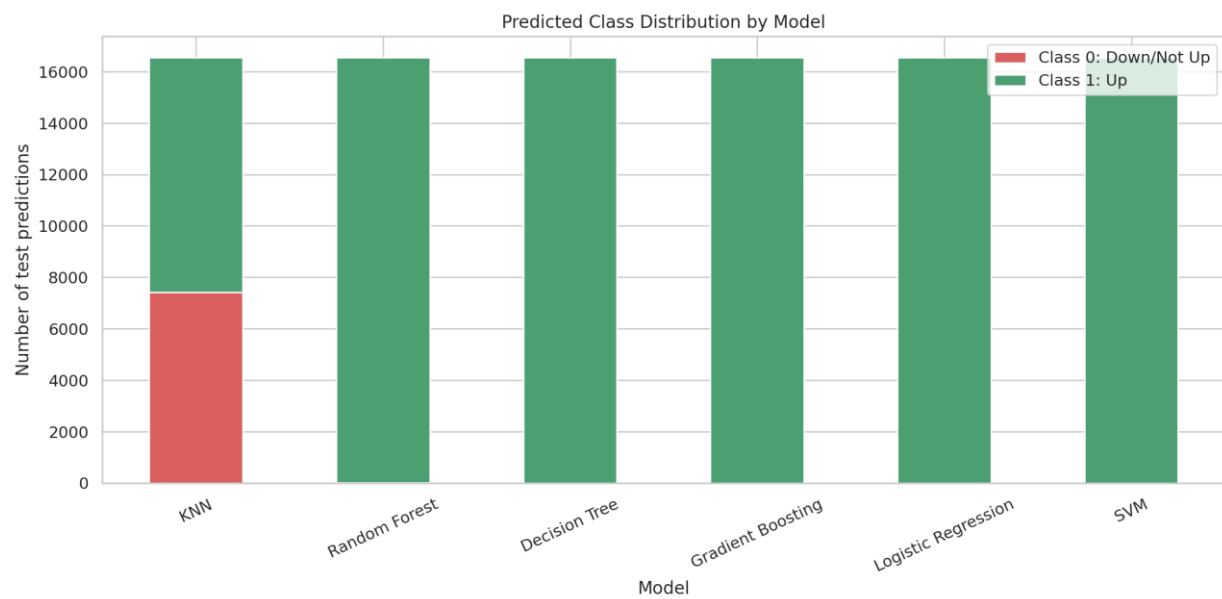


Figure 3 Predicted class distribution by model

4.3 Confusion Matrix Comparison

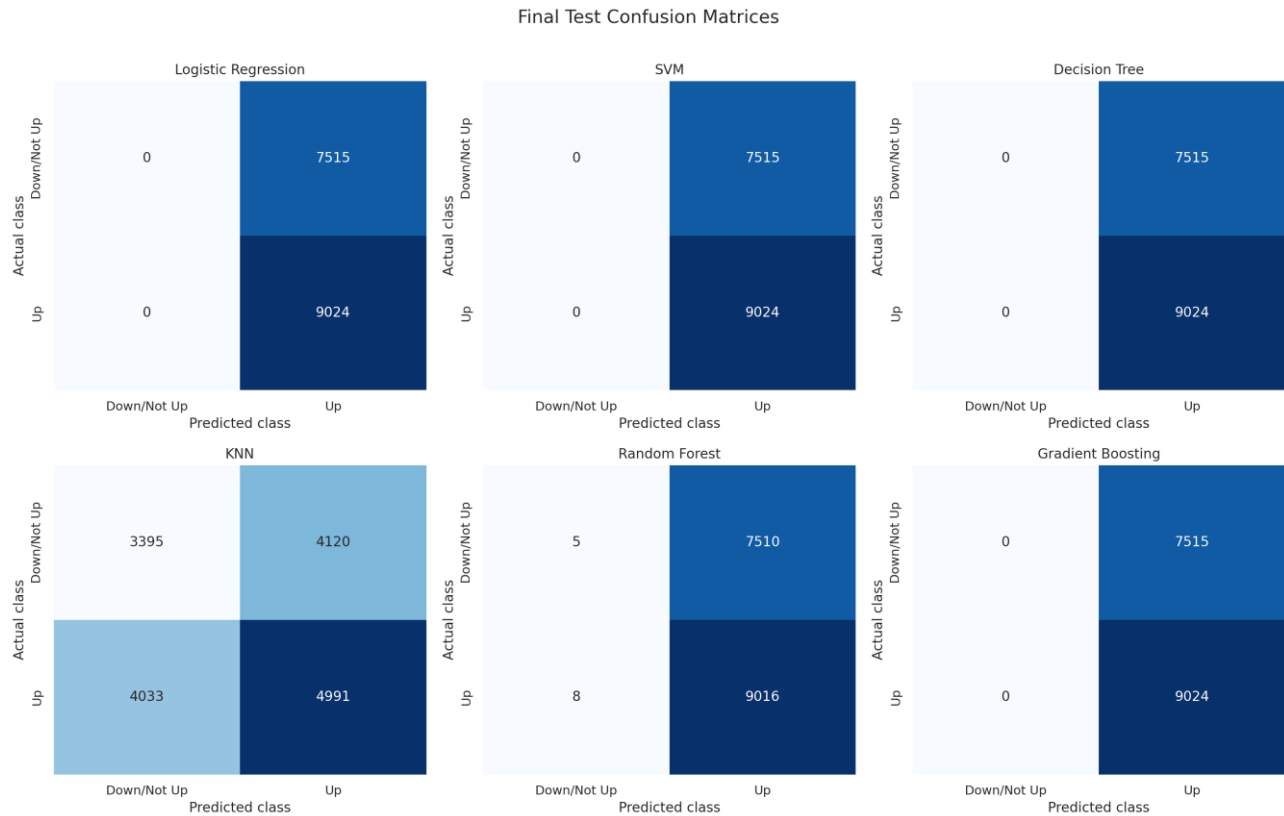


Figure 4 Confusion matrices for the six final test evaluations

The confusion matrices reveal that several models failed to recognize Class 0. Their perfect Class 1 recall was achieved at the cost of zero or near-zero specificity. KNN made substantial predictions for both classes and therefore provided a more meaningful, though still weak, attempt to separate downward and upward movements.

4.4 Five Fold Cross Validation

Model	Mean CV Accuracy	Mean CV F1	Mean CV ROC AUC
Gradient Boosting	0.5294	0.6578	0.5312
Decision Tree	0.5285	0.6421	0.5261
Random Forest	0.5282	0.6431	0.5261
SVM	0.5269	0.6840	0.5168
Logistic Regression	0.5269	0.6840	0.5168
KNN	0.5130	0.5514	0.5108

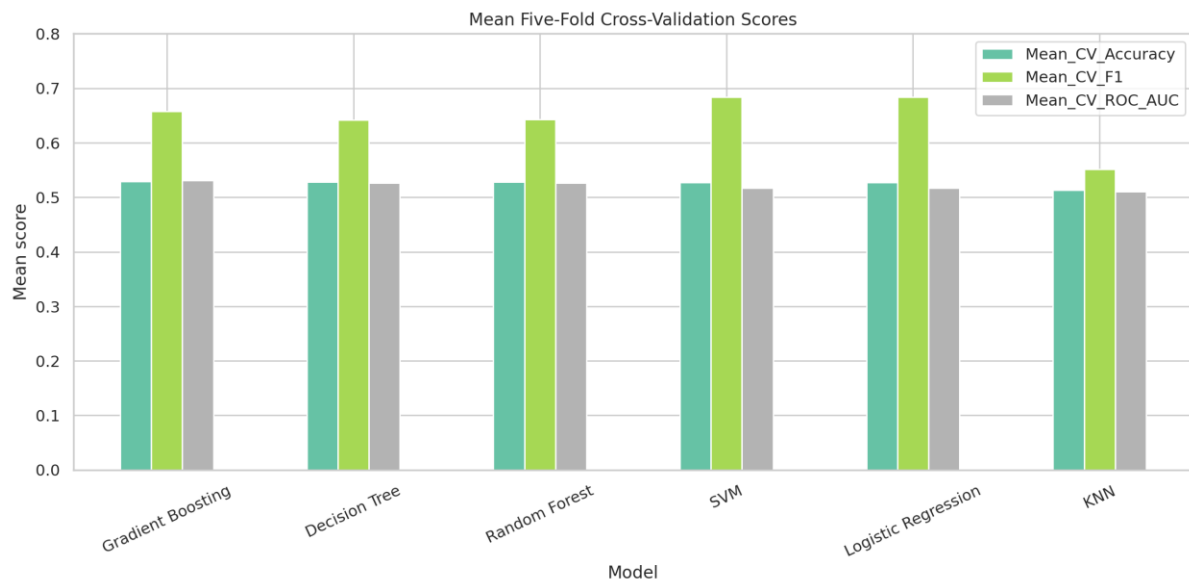


Figure 5 Mean five-fold cross-validation scores

4.5 Final Model Selection

Primary model K Nearest Neighbors. KNN was selected because it produced substantial predictions for both target classes and achieved the highest test ROC AUC and balanced accuracy among the six experiments. It avoided complete collapse to the majority class.

Secondary model Decision Tree. Decision Tree was retained as an interpretable comparison model because its rules and feature importance are easy to explain and it achieved the second-highest test ROC AUC. Its major limitation is that it predicted only Class 1 on the final test set.

The selection does not imply that either model is ready for real financial decision making. KNN achieved only 50.70 percent accuracy and approximately 0.5086 ROC AUC, which is close to random discrimination.

5 Prediction Demonstration

Predictions were generated for 16,539 unseen test observations. Actual_Target is the known test answer, Predicted_Target is the model output, Probability_Up is the score or probability for Class 1, and Prediction_Correct indicates whether the predicted class matched the actual class.

5.1 KNN Sample Predictions

Actual Target	Predicted Target	Probability Up	Correct
1	1	0.6129	True
1	1	0.5161	True
0	0	0.4839	True
0	1	0.5806	False
1	1	0.5484	True
1	1	0.6129	True
1	1	0.6774	True
1	0	0.4516	False
1	1	0.7742	True
1	1	0.5484	True

5.2 Decision Tree Sample Predictions

Actual Target	Predicted Target	Probability Up	Correct
1	1	0.6329	True
1	1	0.5852	True
0	1	0.5754	False
0	1	0.5193	False
1	1	0.5193	True
1	1	0.5193	True
1	1	0.5193	True
1	1	0.5193	True
1	1	0.5193	True
1	1	0.5193	True

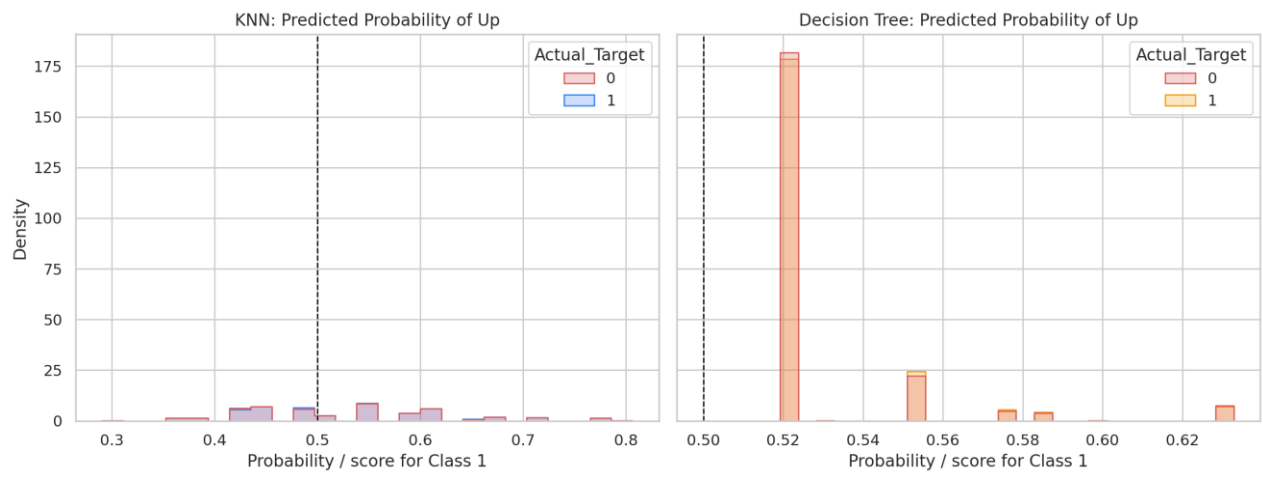


Figure 6 Probability distributions for KNN and Decision Tree

6 Discussion Limitations and Future Work

6.1 Interpretation of Results

All six models demonstrated weak separation between the two target classes. ROC AUC values remained close to 0.50, meaning their ranking ability was approximately random. The apparently stronger accuracy, recall and F1 scores of several models were caused by predicting the majority Class 1 rather than by learning reliable class boundaries.

Low performance is plausible for next-day stock direction because market movements are noisy and influenced by information not contained in the selected features. Hyperparameter tuning can improve how a model learns from available information, but it cannot create predictive information that is absent from the inputs.

6.2 Main Limitations

- The selected five features omitted direct historical-price and return information.
- News, economic announcements, interest rates, market sentiment and political events were unavailable.
- Class proportions changed moderately between training and testing periods.
- F1-based tuning encouraged majority-class prediction in several models.
- Model results were aggregated across different market indices whose behaviour may vary.

6.3 Recommended Future Improvements

- Create lagged returns and previous-day price-change features.
- Add moving averages, momentum indicators and rolling volatility.
- Evaluate index-specific models rather than only one combined model.
- Use class-sensitive learning and tune the decision threshold using validation data.
- Apply walk-forward validation to represent repeated real-world forecasting.
- Include external economic or sentiment data where permitted by the assessment rules.

7 Conclusion

The project completed the required dataset understanding, preprocessing, exploratory analysis, multi-model implementation, hyperparameter tuning, five-fold cross-validation, final evaluation and model comparison. Six supervised classifiers were assessed on the same chronological test partition.

K Nearest Neighbors was selected as the most meaningful observed performer because it predicted both target classes and achieved the highest ROC AUC and balanced accuracy. Decision Tree was kept as a secondary interpretable model. Nevertheless, overall performance remained weak, and the evidence does not support reliable next-day stock-direction prediction with the selected features. This honest negative finding is a valid project outcome and identifies clear directions for future improvement.

